

# **GenieMart**

## **Project Team:**

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# Project Definition

## Statement

University students are always shopping for furniture, clothes, textbooks and other materials. Platforms like Facebook Marketplace and Craigslist that aren't directly geared towards college communities are prone to more scams, unreliable sellers and safety hazards. GenieMart will be geared only towards UNCG students/staff and helping the UNCG community. A platform only accessible to the UNCG community will help ensure safer transactions, more reliable sellers being that the sellers are known to be part of the UNCG community and will help give back to the people of the university. With GenieMart, any UNCG student or person who owns an active UNCG email will be able to put items up for sale that other UNCG members can look at, purchase, comment and favorite. GenieMart will hopefully be a helpful and useful tool for UNCG students and staff to be able to give back to one another in a safer and more reliable way. GenieMart will be a mobile app which helps with convenience and quickness. The architecture of GenieMart will consist of React Native frontend (Expo), Python backend, and a PostgreSQL database.

## Teamwork Division

|                         |                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Eduardo Herrera-Barraza | <ul style="list-style-type: none"><li>• Document Maintainer</li><li>• Backend(First Iteration)</li><li>• Frontend(UI)<ul style="list-style-type: none"><li>▸ User interactive views (splash Screen, login/signup, profile settings)</li><li>▸ Admin interactive views</li></ul></li><li>• Procurement of storage services</li></ul> |
| Lily Nguyen             | <ul style="list-style-type: none"><li>• Document Co-Maintainer</li><li>• Marketing Strategy</li><li>• Backend(First Iteration)</li><li>• Frontend(UI)<ul style="list-style-type: none"><li>▸ User and Hardware interactive View (Main Listings page, More Info, and Listings Creator)</li></ul></li></ul>                           |
| Jared Martinez-Sanchez  | <ul style="list-style-type: none"><li>• Backend(Major Revamp and Final Iteration)</li><li>• Frontend(UI)<ul style="list-style-type: none"><li>▸ Interactive Messaging (API calls based Messaging)</li><li>▸ Major UI Cleanup</li></ul></li><li>• Email Verification system setup</li></ul>                                          |

### ! Important

We took a highly collaborative approach to our work loads everyone played a roll in all aspects of development all contributions outlined are the significant individual contributions!

## Source Code

[Source Code Here. Click Me!](#)

## Video Demo Link

[Demo Video Here. Click Me!](#)

# Project Requirements Analysis

## Functionality

Users will be able to create an account with their uncg email, create listings for items, users can engage sellers through an in-app message board, and view and like listings.

## UI Usability

Cross-platform mobile application, e-commerce type for item listings, and instant messaging for users and listers to engage, and a user dashboard (role-based features)

## Security

GenieMart is strictly for UNCG students; security is ensured by only allowing @uncg.edu domain emails to create an account, on top of verifying student status by having users upload pictures of their student identification card before buying and listing features are granted. Student identification cards are stored temporarily, and then the system will delete them after the verification process occurs.

# Project Specification

## Focus/Domain/Area

The focus of our project is to provide the UNCG community with an online marketplace that enables students to safely buy and sell items within their campus community using university email authentication.

## Libraries/Framework/Development

### Backend Framework

**Programming Language:** Python

### Web framework

- FastApi → Is the base of backend interaction, handles model, schema, CRUD, and routing (API endpoints creation and accessing).
- SQLAlchemy/Alembic → Type-safe queries and data migration that assisted through the iteration of our backend.
- Uvicorn → Local development Server.
- Render → Hosting platform for production server.

### Data Persistence/Object & File Storage

- Dockerized Postgresql Database → Development Database

- Neon Serverless Postgresql → Production database goes hand in hand with Render, our chosen hosting platform for backend services.
- Cloudflare R2 → Stores and handles the processing of images uploaded by users.

### **Authentication**

- JWT(JSON Web Tokens) → primary authenticator, if you DO NOT have one generated by a login you simply cannot access the app
- PWDLIB → Password hasher, storing the literal password in a database is not secure, therefore all passwords must be hashed.
- PyJWT → JSON web token encoder and decoder.

#### **Note**

Our application is role-based; there are general users i.e. sellers and buyers, and there is an administrator role which is taken up by the dev team. Role-based login is handled using JSON Web Tokens.

### **Frontend Framework**

**Programming Language:** Typescript

- React Native → Multiplatform UI framework for mobile development.
- Expo → Framework that provided mobile development tools.
  - Camera roll access and camera access provided by the expo framework allow for image upload
- Expo-Router → File-based routing framework, provided our tab based in app navigation.

### **Platform**

- Mobile Application → IOS and Android compatible

### **Genre**

- Online Multi-Vender Meetup Marketplace.

# System Design

## Entities/Objects

| Category   |
|------------|
| id         |
| name       |
| parent_id  |
| created_at |

| Favorite   |
|------------|
| id         |
| user_id    |
| listing_id |
| created_at |

| Listing Image |
|---------------|
| id            |
| listing_id    |
| image_url     |
| is_primary    |
| created_at    |

| Message      |
|--------------|
| id           |
| thread_id    |
| message_user |
| body         |
| created_at   |

| Email Verification Token |
|--------------------------|
| id                       |
| user_id                  |
| token                    |
| expires_at               |
| used_at                  |
| created_at               |

| Listing     |
|-------------|
| id          |
| title       |
| description |
| price       |
| seller_id   |
| created_id  |
| updated_at  |
| condition   |
| category_id |
| status      |
| location    |

| Listing Offer |
|---------------|
| id            |
| listing_id    |
| buyer_id      |
| offer_price   |
| created_at    |

| Message Thread |
|----------------|
| id             |
| seller_id      |
| buyer_id       |
| listing_id     |
| created_at     |

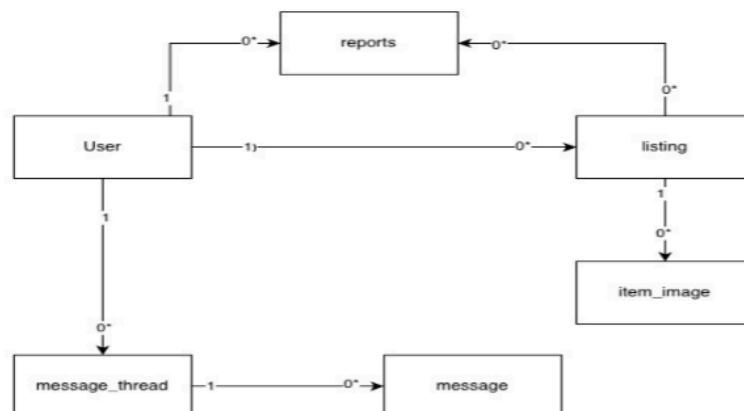
| Notification  |
|---------------|
| id            |
| user_id       |
| actor_user_id |
| thread_id     |
| listing_id    |
| type          |
| content       |
| is_read       |
| created_at    |

| Report      |
|-------------|
| id          |
| reporter_id |
| listing_id  |
| reason      |
| created_at  |

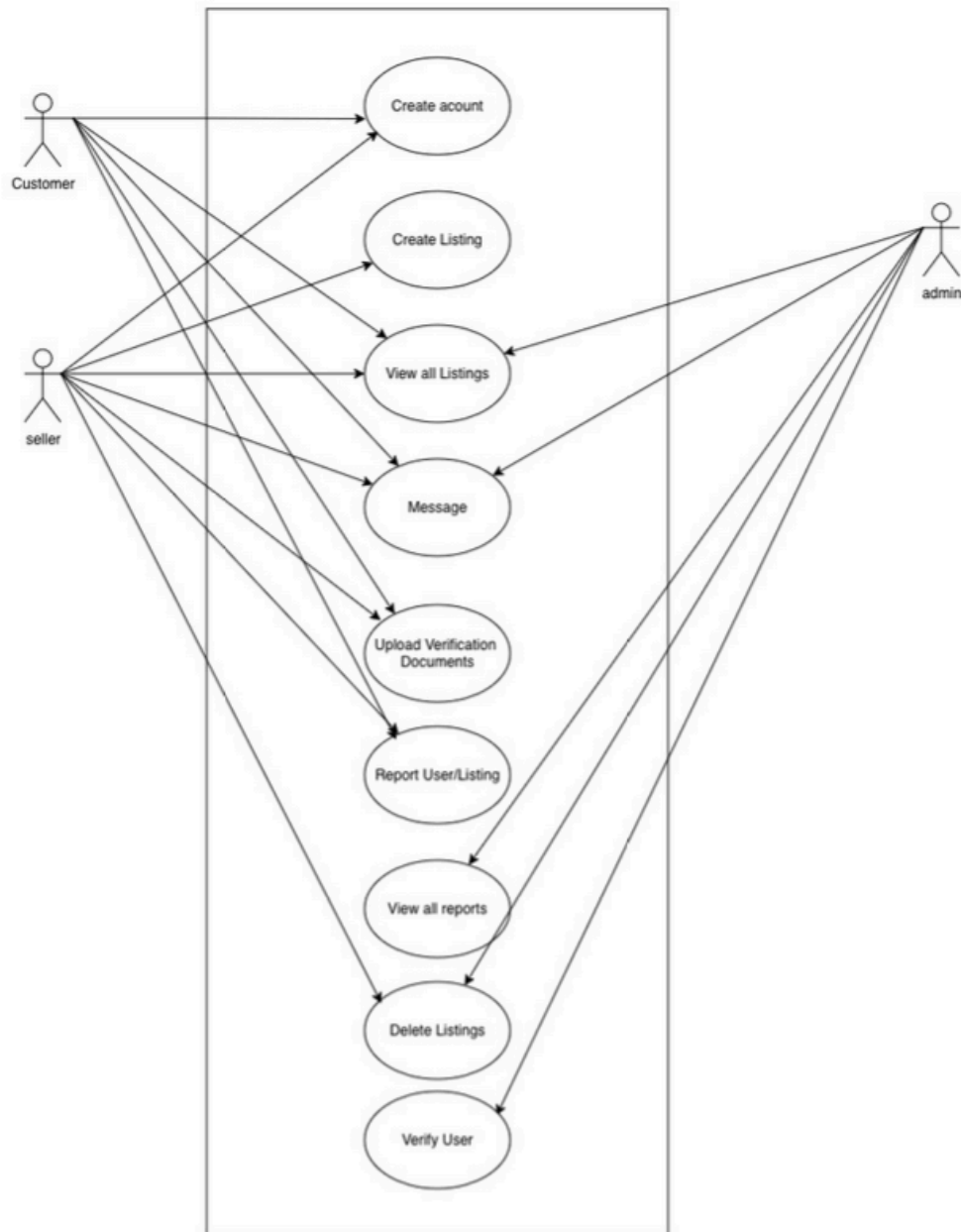
| Review           |
|------------------|
| id               |
| reviewer_id      |
| reviewer_user_id |
| rating           |
| comment          |
| created_at       |

| User              |
|-------------------|
| id                |
| username          |
| profile_image_url |
| is_admin          |
| is_banned         |
| school_email      |
| phone_number      |
| is_email_verified |
| email_verified_at |
| phone_verified_at |
| password_hash     |
| created_at        |
| updated_at        |

## Entity Relations



## Use Cases



## Evaluation

GenieMart has a typical usage flow, a user launches the app on their preferred platform (IOS or Android) they are met with a splash that asks them to click to continue, then they are presented with two options; create an account which will launch a creation form at the end of which will instruct the user to confirm their email before they are allowed inside the market place, the other option is to directly login if they already have an account. The user is then presented with an index that serves the listings browser with



an inner search engine, the next view is the “create listing” tab which allows a seller to create their own item listings, the user also has the options to see their listings in a separate tab which also allows for editing and deleting, the last view the user has is a profile tab which has quick access to liked items, profile image changing, and general profile information.

## Discussion

GenieMart is currently in its first iteration, the app is function and the purpose of the app is complete. GenieMart’s dev team has plans to integrate the GoogleMapsAPI/ SDK to provide with a mapping of listings around the University of North Carolina at Greensboro’s community on a map that users can navigate to buy items near their relative location. The dev team is also poised to integrate payment features to allow sellers to be paid for their items in app using a service such as paypal moving from a cash purchasing only platform to a multipayment platform while maintaining the in-person meetup to retrieve the item purchased.

## Would you like to try it out?

1. Clone the project on your machine.
2. Make sure you have Node and Python installed on your system.
3. Move into the frontend directory like so:

```
cd frontend
```

4. Install the npm packages like so :

```
npm install
```

5. Now that all packages are installed you may run the frontend and follow the CLI to launch the app like so :

```
npx expo start
```

### Note

You do not need to run the backend as it is hosted on the render platform!

 **Warning**

You must have an IOS/Andriod emulator installed to run the frontend on a laptop or PC. You can also use your own celular device to run it but you will need to install the expo app from the App Store or Google Play Store and follow the CLI to pipe the output to the device.